

# Rolling deployment upgrade

Previous single-instance command compared with the new three-instance, stop-on-failure workflow.

READY TO USE  
TESTED WITHOUT  
DEPLOYING

## Executive result

| One command                       | Availability                                                                                    | Failure policy                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Deploys EC2 #1, then #2, then #3. | Only one instance is restarted at a time; the next host is untouched until verification passes. | The failed host rolls back to its own previous commit. The sequence stops immediately. |

## Before and after

| Area         | Previous version                                                    | New version                                                                              |
|--------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Targets      | One fixed EC2 address.                                              | Three fixed production EC2s in required order.                                           |
| Commit       | Each invocation followed that server's current <b>origin/main</b> . | Pins one 40-character SHA from <b>origin/main</b> ; every host must still resolve to it. |
| Verification | Health, courses, PM2, port, and commit checks on one host.          | Adds <b>/api/ready</b> ; all checks complete before the next host begins.                |
| Failure      | Single-host rollback only.                                          | Same tested rollback on the failed host, followed by immediate rolling-stop.             |
| Concurrency  | Per-server <b>flock</b> only.                                       | Atomic Mac-side rolling lock plus existing per-server <b>flock</b> .                     |
| Logs         | Remote per-instance log.                                            | Remote logs retained; matching local log created for each instance.                      |

## Files changed

|                                                                                        |                                                                                       |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>/Users/kratik/deploy-skillomate-production.sh</b><br>Installed command entry point. | <b>deployment/deploy-skillomate-production.sh</b><br>Three-host rolling orchestrator. |
| <b>deployment/deploy-skillomate.sh</b><br>Pinned target and readiness-aware rollback.  | <b>deployment/test-rolling-deploy.sh</b><br>Isolated safety tests.                    |
| <b>deployment/README.md</b><br>Operating and recovery documentation.                   | <b>output/pdf/...upgrade-report.pdf</b><br>This verified comparison report.           |

## Deployment flow

- Acquire the local rolling-deployment lock. A concurrent command exits with status 75 before SSH activity.
- Resolve and pin the full **origin/main** SHA from EC2 #1.
- For each EC2 in order, stream the repository-owned deployer over SSH. No application files or **.env** are copied from the Mac.
- On that host: acquire its **flock**, validate repository and ignored **.env**, fetch GitHub, require pinned SHA equals **origin/main**, and reset to that SHA.
- Build the frontend, run backend **npm ci --omit=dev**, restart **skillomate\_backend --update-env**, and run **pm2 save**.
- Require health and readiness HTTP success, valid courses JSON, PM2 online, port 3000 listening, and HEAD equal to the pinned **origin/main** SHA.
- Write the instance result to local and remote logs. Only then start the next EC2.

## Rollback flow

| Trigger                                                 | Action                                                                                                                                | Required proof                                                                                  |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Any failure after the commit switch on the current EC2. | Reset that EC2 only to its captured previous commit; reinstall frontend/backend dependencies; rebuild frontend; restart and save PM2. | Health and readiness succeed, courses JSON is valid, PM2 is online, and port 3000 is listening. |
| Preflight failure before switching commits.             | No rollback is needed because production files were not changed.                                                                      | The orchestrator stops; all later EC2s remain untouched.                                        |

## Safety guarantees

- The existing production **.env** must exist, be mode 0600, be Git-ignored, and remain untracked. It is never copied or generated.
- Each host keeps its own previous commit for rollback; earlier successful hosts are not rolled back when a later host fails.
- A moving **origin/main** fails closed instead of producing mixed commits across instances.
- This tooling update did not invoke the deployment command and did not modify AWS, application code on EC2, MongoDB, Redis, or production data.

## Verification performed

| Check                                                          | Result                                   |
|----------------------------------------------------------------|------------------------------------------|
| Bash syntax for orchestrator, per-instance deployer, and tests | PASS                                     |
| Three-host success order                                       | PASS - #1, #2, #3                        |
| Failure on EC2 #2                                              | PASS - exit propagated; EC2 #3 untouched |
| Concurrent invocation lock                                     | PASS - rejected before SSH               |
| Existing rollback reset behavior in a temporary Git repository | PASS - prior commit restored             |
| Production deployment                                          | NOT RUN (explicitly prohibited)          |